The Delighful Dishes of Combinatorics

Problem 1

How many 6-digit numbers have strictly-decreasing digits when read from left to right?

Solution 1

A 6-digit strictly decreasing number is uniquely determined by choosing 6 distinct digits from $\{0, 1, \ldots, 9\}$, since once the digits are chosen there is exactly one way to arrange them in strictly decreasing order.
Therefore the number of such 6-digit numbers is:

$$\binom{10}{6} = 210.$$

Note that we need not worry about leading zeros, since if 0 is among the chosen digits it will always appear last in the decreasing arrangement.

Problem 2

How many 6-digit numbers have non-decreasing digits when read from left to right?
Hint: Consider a similar approch to stars and bars.

Solution 2

To enumerate a number we, first consider listing the digits 1 to 9 in a line. For every digit in our number, we place a single line after the value of that digit in our line. For example, we can represent:

$$(145557) \to 1|234|5|||67|89.$$

Note that there will never be a line before the 1, so the digit 1 can be ignored:

$$|234|5|||67|89.$$

Each different placement of the bars uniquely represents a unique valid number, and the number of placements can be found using stars and bars by letting the digits $2, \ldots, 9$ be dots and the 6 digits of the number be lines. Thus, the solution is:

$$\binom{8+6}{6} = 3003.$$

Problem 3

Suppose that $k, n \in \mathbb{Z}$.

(a) How many binary numbers of length at most n exist with exactly k 1's?

(b) How many binary numbers of length at most n exist with at least k 1's? (Closed form not expected here.)

Solution 3

We solve the parts in the order they were presented.

(a) The length of the string could be $k, k+1, \ldots, n$. For a fixed length m, there are $\binom{m}{k}$ choices for the positions of the 1's. Therefore, the total number is

$$\sum_{m=k}^{n} \binom{m}{k} = \binom{n+1}{k+1},$$

by the hockey stick identity.

(b) Similarly, the length of the string could be $k, k+1, \ldots, n$. For a fixed length m, the number of 1's could be $k, k+1, \ldots, m$. Suppose there are r 1's; then the number of such strings is

$$\binom{m}{r}.$$

Summing over all possible r and m, the total number is

$$\sum_{m=k}^{n} \sum_{r=k}^{m} \binom{m}{r}.$$

Problem 4

King Justin IV is climbing pizza-decorated n stairs of his Syldavian Palace with steps of 1 or 2. How many unique combinations of steps can he take (n one step, $n-2$ one step and 1 two steps, etc)?

Solution 4

Intuitively this is an application of the Fibonacci sequence as Justin's possible step combinations come from having already climbed $n-1$ steps or $n-2$ steps from where he just need to takes 1 more step. Hence if we denote E as an expected number of steps, we are asked to solve $E(n) = E(n-1) + E(n-2)$ with initial conditions $E(0) = 1$ as one way to be at the bottom and $E(1) = 1$ as one way to climb one step. The $E(n)$ equation can be solved by discrete time system ODE Method $x^2 - x - 1 = 0$ which yields $\frac{1}{2}(1 \pm \sqrt{5})$ or the golden ratio as roots, causing our general solution to be $E(n) = A_0 \cdot (\frac{1}{2}(1-\sqrt{5}))^n + A_1 \cdot (\frac{1}{2}(1-\sqrt{5}))^n$ where once substituting the initial conditions, we reach our answer which is Binet's formula.

Problem 5

Prove that if m, n are positive integers and $1 \leq k \leq n$, then

$$\binom{n}{0}\binom{m}{k} + \binom{n}{1}\binom{m}{k-1} + \cdots + \binom{n}{k}\binom{m}{0} = \binom{m+n}{k}.$$

Hint: Consider selecting from a set of people of size $n+m$ that has n male and m female.

Solution 5

Consider a group of n men and m women. The right-hand side $\binom{m+n}{k}$ counts the number of ways to form a committee of k people from all $m+n$ individuals.

For the left-hand side, we partition this selection by the number of men chosen. If exactly i men are selected from the n available, then exactly $k-i$ women must be selected from the m available, giving $\binom{n}{i}\binom{m}{k-i}$ ways. Since i can range from 0 to k, summing over all cases gives the left-hand side.

Both sides count the same quantity, so they are equal.

Alternative solutions include considering the coefficients of x^k in $(1+x)^m(1+x)^n = (1+x)^{m+n}$.

Problem 6

Suppose that $n \geq m$, Prove that

$$\sum_{k=m}^{n} \binom{n}{k}\binom{k}{m} = 2^{n-m}\binom{n}{m}.$$

Hint: Consider the number of subsets of sets of size n.

Solution 6

We count the number of pairs (A, B) where $A \subseteq B \subseteq \{1, 2, \ldots, n\}$ with $|A| = m$.

RHS: First choose A of size m from $\{1, \ldots, n\}$ in $\binom{n}{m}$ ways. Each of the remaining $n-m$ elements independently either belongs to B or not, giving 2^{n-m} choices for B. This yields $2^{n-m}\binom{n}{m}$ pairs.

LHS: We instead construct the pair by choosing B first. If $|B| = k$, then k must satisfy $m \leq k \leq n$. There are $\binom{n}{k}$ ways to choose B, and then $\binom{k}{m}$ ways to choose $A \subseteq B$ with $|A| = m$. Summing over all valid k gives:

$$\sum_{k=m}^{n} \binom{n}{k}\binom{k}{m}.$$

Since both sides count the same set of pairs, we conclude

$$\sum_{k=m}^{n} \binom{n}{k}\binom{k}{m} = 2^{n-m}\binom{n}{m}.$$

Problem 7

At a research station on The Skeld, Captain Red has $(n+1)$ beans, n Crewmates of distinct colors and one **Morphling** who shapeshifts to mimic the color of anyone it spends the night with. Captain Red selects 4 beans at random and locks them in Security overnight. The next day, he must split them into two separate pairs such that **no pair contains two beans of the same color**. Prove that the number of possible color pairings is:

$$3\binom{n+1}{4} = \binom{\binom{n}{2}}{2}.$$

Solution 7

We count the number of valid color pairings in two ways.

LHS: First, select the "Sus Squad" of 4 beans in $\binom{n+1}{4}$ ways.

- If the Morphling is *not* chosen, we have 4 distinct colors. There are exactly 3 ways to partition 4 colors into two pairs.
- If the Morphling *is* chosen, there are 3 other crewmate colors present. The Morphling mimics one of them (3 choices). To avoid a color-clash, the "twins" must be separated into different pairs. For each mimic choice, there is exactly 1 valid way to pair the remaining beans. This again yields 3 possible color pairings.

Summing these cases gives $3\binom{n+1}{4}$.

RHS: We instead count by choosing the color combinations directly from the crew's palette. There are n colors available, so there are $\binom{n}{2}$ possible unique color pairings. To form two distinct pairs for the task, we simply choose 2 different color combinations from this "pool." This gives:

$$\binom{\binom{n}{2}}{2}.$$

Since both sides count the same total number of color-distinct task pairings, the identity holds.

Problem 8

Consider the set $2^{[n]}$, where $[n] := \{1, 2, \ldots, n\}$ is the index set of n elements. Prove that the minimum number of chains $\mathcal{C}_i = \{C_1, C_2, \ldots, C_m\}$, where

$$C_1 \subseteq C_2 \subseteq \cdots \subseteq C_m,$$

required to cover $2^{[n]}$ is at least $\binom{n}{\lfloor \frac{n}{2} \rfloor}$.

Hint: If $A, B \subseteq [n]$ are such that $|A| = |B|$, then we cannot have $A \subseteq B$ or $B \subseteq A$.

Solution 8

We work in $(2^{[n]}, \subseteq)$, which is a partially ordered set.
In this partial order, each $\mathcal{C}_i$ is a chain. Hence, it suffices to show that there exists an antichain of size at least k, by Dilworth's theorem. We can do this by considering all subsets of size $\lfloor \frac{n}{2} \rfloor$. This forms an antichain, since sets of the same size cannot be subsets of one another.
Note that it is also true that $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ is sufficient. See here. One way to prove this is to construct a chain decomposition that intersects the middle layer (the set of elements with cardinality $\frac{n}{2}$), using symmetry (see here, here).

Problem 9

Let S be a set of n^2+1 points in $\mathbb{R}^2$. Prove that there is a sequence of $n+1$ points

$$(x_1, y_1), \ldots, (x_{n+1}, y_{n+1}) \in S \text{ such that } x_1 \leq x_2 \leq \cdots \leq x_{n+1},$$

and either

$$y_1 \geq y_2 \geq \cdots \geq y_{n+1} \text{ or } y_1 \leq y_2 \leq \cdots \leq y_{n+1}.$$

Solution 9

Order the points based on their x-coordinates and apply the Erdős–Szekeres Theorem (proved using Dilworth's theorem in the slides) with $r = s = n$.

Problem 10

A 6 bead white coloured necklace is being coloured into blues, blacks, or browns. Given all beads are coloured, how many unique combinations exist?

Solution 10

Since there are 6 beads, there are 3^6 ways to colour.

Rotating across the necklace by $1, 5$ beads only has one symmetry when they are all the same colour, hence 3 ways for $1, 5$ rotations by each.

Rotating by $2, 4$ beads only has symmetries if every second bead is the same colour, hence 3^2 ways for $2, 4$ rotations by each.

Similar case for 3 beads, 3^3 ways.

With reflection, symmetries exist with "palindromes" and these palindromes depend on the selection of colouring the first three consecutive beads, so more symmetries created in 3^3 ways.

Now $1, 4$ and reflection create 3^4 for each for combination.

Similarly for $2, 4$ and reflection, 3^3 for each.

For 3 beads and reflection, 3^4 ways.

So the total number of symmetries are $(3^6+2\cdot3+2\cdot3^2+3^3)+(3^3+2\cdot3^4+2\cdot3^3+3^4) = 1104$.

Our $G =$
$\{1 \text{ rotate+reflect}, 5 \text{ rotate+reflect}, 2 \text{ rotate}, 4 \text{ rotate+reflect}, 3 \text{ rotate+reflect}, \text{reflect/rotate}\}$
So $|G| = 12$. Hence by Burnside's Lemma, there are $1104/12 = 92$ necklaces.

Problem 11

An 8 digit binary sequence is constructed by choosing 0s or 1s for each digit. How many unique binary digit sequences can be created, noting that rotating (shifting last digit to the front) can create symmetries?

Solution 11

Application of Burnside's Lemma. $|G| = 8$ as sequence can be rotated 1 to 8 beads (no real reflection like with beads) across uniquely. Now the permutations of a sequence including symmetries as 2^8 plus 2 (all zeroes or ones rotated once) plus 2^2 (all zeroes or ones or 01 or 10 patterned rotated twice) + 2 (all zeroes or ones rotated thrice) + 2^4 (all palindromes regarding rotation four times) plus $2+4+2$ (rotation five times, six times, seven times same as once, twice, thrice) which sums to $256+2+4+2+16+2+4+2 = 288$, before dividing by $|G|$, to yield $288/8 = 36$.

Problem 12

Master Chen is hosting a round-robin tournament between 4 players. In how many ways can he host the tournament, which is a collection of matches, where a match is exclusively between two players that are different from each other and each players plays each other regardless of previous matches?

Solution 12

We apply Burnside's Lemma. $|G| = 4! = 24$, as it is the number of permutations in ordering the 4 players.

In terms of symmetries, $2^{{}^4C_2} = 64$ is the main way to chose tournaments, then there are ${}^4C_2 = 6$ ways to create symmetries of 2 player matches which can be chosen in $2^{1+{}^3C_2} = 16$ ways. Then 3 ways to host matches of 3, chosen in $2^{1+{}^3C_1} = 16$ ways. Then for 4 player match symmetries, there are 4 ways to order them in terms of playing with $2^3 = 8$. Now additionally, rotating, of which there are ${}^4C_2 = 6$ of swapping between players and 4 ways to chose the player to swap from. This makes the total number of permutations $64 + 6\cdot16 + 3\cdot16 + +8\cdot4 + 6\cdot4 = 264$.

Divding by $|G|$, we have the number of unique ways as $264/24 = 11$.